

PrismPrune: Decoupling Saliency and Diversity in Attention for Efficient Visual Token Pruning in VLMs

Ziniu Liu^{1,2} Shuheng Zhou² Mingqing Liu¹ Hao Deng^{1*} Huijia Zhu^{2*}

¹Tongji University ²Ant Group

denghao1984@tongji.edu.cn liuziniu.lzn@antgroup.com

Abstract

Vision-Language Models (VLMs) have achieved unprecedented performance by adopting high-resolution inputs, but at the cost of a quadratic surge in visual tokens that creates a severe computational bottleneck. Existing token pruning methods attempt to mitigate this, but struggle to balance saliency and diversity efficiently. Saliency-focused approaches discard vital context and visual diversity, while methods that incorporate diversity risk sacrificing essential salient tokens and are sensitive to initialization. This paper introduces PrismPrune, a novel and efficient pruning framework that resolves this dilemma. Our core insight is that the signals for saliency and diversity are already intrinsically decoupled within the multi-head attention mechanism. PrismPrune operationalizes this insight by decomposing the attention signal into its constituent components. First, it preserves the most salient tokens identified by the mean attention score. Second, it captures diverse tokens by leveraging attention variance within an efficient “filter-then-sample” strategy, which first filters for a high-potential candidate pool before performing diversity sampling exclusively on this subset. This hybrid approach thereby maintains a rich representation of both salient and contextual information. Extensive experiments demonstrate state-of-the-art results. On LLaVA-1.5, we prune 77.8% of tokens with zero performance loss. For the high-resolution LLaVA-NeXT, we reduce pre-filling time by 10x and FLOPS by 13x while retaining over 91% of the original performance, significantly outperforming existing methods.

1. Introduction

VLMs[4, 10, 23, 28, 39] have revolutionized our comprehension of the multimodal world by synergizing the profound reasoning capabilities of Large Language Models (LLMs)[1, 3, 40] with the perceptual acuity of computer vision. A definitive technological trajectory in

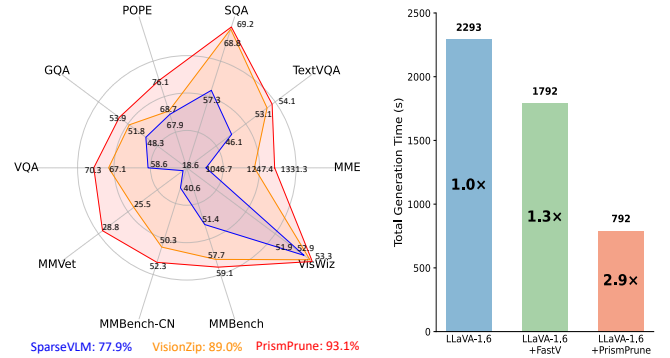


Figure 1. **PrismPrune significantly outperforms SOTA efficient VLM methods, such as SparseVLM and VisionZip.** As shown with LLaVA-1.5, it retains 93.1% of the original performance across 10 benchmarks while using a mere 5% of the visual tokens. Furthermore, when applied to LLaVA-1.6, it accelerates generation time by 2.9x.

the pursuit of deeper visual understanding is the adoption of higher-resolution inputs to capture finer image details. This strategy, exemplified by pioneering models such as LLaVA-NeXT[30] and InternVL 1.5[9], has led to unprecedented performance on tasks demanding intricate visual understanding, including complex visual question answering[19], fine-grained recognition[18], and optical character recognition[22]. However, this performance enhancement is accompanied by a formidable challenge: a dramatic proliferation in the number of visual tokens.

These excessively long visual tokens consume a significant amount of memory and computation in the entire VLM. In response to this challenge, visual token reduction[6, 8, 16, 51] has emerged as a promising technique. This method usually first defines the importance of each token, and then prunes the most unimportant tokens during inference. However, a fundamental question remains: *how can we efficiently and accurately define and identify this set of “important” tokens?*

Early approaches predominantly relied on singular met-

*Corresponding author



Figure 2. **Visualization of Decoupled Attention Signals.** Mean attention focuses on the primary subjects (oranges), while the attention variance heuristic successfully identifies diverse contextual elements like background details and textual information.

rics, such as attention scores[17, 47, 58], to capture saliency. However, using attention scores in LLMs exhibits poor compatibility with efficient implementations like FlashAttention[11] and can be biased towards text-relevant tokens rather than those that are visually critical[56]. Subsequent research[43, 49, 50] attempted to address this by incorporating diversity metrics, for instance, through bipartite matching[6] or minimax sampling[2]. While effective, these methods face their own challenges: they risk discarding salient tokens at high pruning rates[35], and their performance can be sensitive to initialization[41]. Moreover, executing these global diversity algorithms across all tokens is computationally expensive. Therefore, the field urgently requires a new framework that is both effective and efficient.

Our work is founded on a key insight: **the signals for capturing saliency and diversity are already intrinsically decoupled and co-exist within the multi-head attention mechanism.** Instead of attempting to balance these two objectives at a high level, we propose to delve into the attention mechanism itself by conceptualizing it as an intrinsic ensemble system. This perspective is powerful: the multi-head diversity has been empirically established [54], and recent theory formally links ensemble diversity to the variance among its members [46]. Specifically, we utilize the mean of attention scores as a direct indicator for saliency, while introducing the variance of attention scores across heads as a novel and powerful heuristic for identifying information-rich tokens that contribute to diversity (see Figure 2 for a visualization).

Motivated by this insight, we introduce PrismPrune named for its ability to decompose the attention signal like a prism. PrismPrune first secures the most visually critical information by preserving a subset of tokens with the highest attention scores, directly addressing the saliency objective. From the remaining tokens, it then performs an efficient “filtering-then-sampling” process by first leveraging our proposed attention variance metric to filter for a candi-

date pool. Subsequently, inspired by DivPrune[2], we execute a Farthest Point Sampling[13] algorithm for diversity sampling on this pool, thereby reducing the computational overhead while preserving high performance. In the training paradigm, to mitigate the slight modality misalignment inevitably introduced by any pruning method, we incorporate a lightweight fine-tuning step. This involves around 2 hours fine-tuning of the projection layer.

We evaluated PrismPrune by applying it to popular VLMs across multiple benchmarks. As shown in Figure 1, the experimental results demonstrate that our training-free version significantly outperforms existing state-of-the-art methods. This superiority is best demonstrated by its remarkable efficiency, as our method can prune 77.8% of the visual tokens in LLaVA-1.5-7B while maintaining 100% of its original performance across 10 different tasks. Furthermore, for models with longer visual token sequences, such as LLaVA-NeXT, our approach achieves a 10-fold reduction in prefilling time and a 13-fold reduction in FLOPS, while still retaining 91.2% of the original performance.

2. Related works

2.1. Vision Language Models

Inspired by the monumental success of LLMs[1, 3, 40] in Natural Language Processing, VLMs[4, 10, 23, 29, 39] have emerged to forge a cross-modal understanding of the physical world. The core objective is to synergize the exceptional reasoning and generative capabilities of LLMs with the perceptual acuity of computer vision. To this end, prevailing VLMs typically employ a modular architecture with three principal components: a vision encoder[36, 55], a large language model, and a modality alignment module.

In the pursuit of enhanced performance, a dominant paradigm has been the adoption of higher-resolution input images. This approach enables models to capture finer visual details, culminating in the superior reasoning abilities demonstrated by state-of-the-art models like LLaVA-NeXT[30], InternVL 1.5[9], and Qwen2.5-VL[5], and has led to leading performance on numerous authoritative benchmarks. However, this strategy comes at a considerable cost: a quadratic surge in the number of visual tokens. This directly results in a sharp escalation of computational costs and a significant increase in inference latency, presenting a formidable challenge to the deployment of VLMs in resource-constrained environments or applications with stringent real-time requirements.

2.2. Token Reduction for VLMs

Visual token reduction techniques accelerate inference by eliminating redundant information, often leveraging textual guidance. This leads to two pathways: attention-based internal pruning[31, 52] and pre-processing feature fusion[25,

44, 57]. In internal pruning, attention scores serve as a proxy for token importance. For instance, FastV[8] prunes low-scoring tokens in the shallower layers of the language model, while SparseVLM[58] utilizes more fine-grained text attention to guide the sparsification process. Moreover, VTW[27] observes that the contribution of visual tokens diminishes in the deeper layers of the LLM, leading to their proposal of removing them entirely. Concurrently, methods like LLaMA-VID[25] and LLaVA-MINI[57] pursue pre-processing fusion, where multimodal information is first fused before the LLM to generate a more compact feature representation. However, subsequent research[56] has revealed inherent drawbacks of relying solely on the LLM’s internal attention, including positional bias and attention scattering. More critically, the requisite access to the attention matrix conflicts with the design of modern efficient inference frameworks such as FlashAttention[11].

Consequently, other research has started to move beyond the attention-based framework to explore the intrinsic properties of visual tokens themselves. For example, DART[45] innovatively posits that token redundancy is a more fundamental criterion for pruning. It achieves efficient and hardware-friendly acceleration by removing tokens that are highly similar to a small set of pivot tokens. DivPrune[2] explicitly formulates this as a max-min optimization problem to enable diversity-driven token selection. G-Prune[21] introduces a novel graph-based perspective, treating tokens as nodes in a graph. By propagating information across the graph structure, it identifies the most representative token subset. VisPruner[56] and VisionZip[50] both employ a hybrid strategy, combining attention-based importance selection with similarity-based diversity preservation.

3. Method

3.1. Preliminary

Vision-Language Model Architecture. Our approach is designed to be integrated into the generic architecture of modern VLMs. Such models are typically composed of three principal components: A *vision encoder*, commonly a Vision Transformer (ViT)[12], which encodes an input image $\mathcal{I}$ into a sequence of visual tokens $\mathcal{X} = \{x_1, x_2, \dots, x_n\}$, where $\mathcal{X} \in \mathbb{R}^{N \times C}$, with N being the number of tokens and C the feature dimensionality. A *LLM*, acting as the cognitive core of the model, which processes the input sequence and generates textual outputs. A *modality alignment module*, which projects the visual tokens from the visual feature space into the LLM’s embedding space.

Vision Transformer and Self-Attention. The internal mechanism of the ViT encoder provides the crucial motivation for our pruning strategy. A ViT processes an image by tokenizing it into a sequence $\mathcal{X}$, which is refined through stacked Transformer[42] layers. For a single attention head,

Algorithm 1 Pseudocode for PrismPrune

```
# X: All visual tokens, shape (N, C)
# A: Attention maps, shape (H, N, N)
# K_sal, K_div: Number of tokens to select
# M: Size of the candidate pool for diversity

# Compute mean attention scores
mean_attn = A.mean(dim=0)
sal_indices = mean_attn.topk(K_sal).indices

# Compute attention variance
var_attn = A.var(dim=0)

# Mask out already selected tokens
var_attn[sal_indices] = -infinity
cand_indices = var_attn.topk(M).indices

# Gather features for the candidate tokens
cand_features = X[cand_indices]

# Select diverse tokens from the pool using FPS
diverse_local_indices = FarthestPointSampling(
    cand_features, K_div) # See Algorithm 2

# Map local indices back to global indices
diverse_indices = cand_indices[
    diverse_local_indices]

# Concatenate indices for final set J
selected_indices = cat([sal_indices,
    diverse_indices])
```

this process yields an attention map $\mathbf{A}_{\text{single}} \in \mathbb{R}^{N \times N}$ reflecting the relationships between tokens:

$$\mathbf{A}_{\text{single}} = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} \right), \quad (1)$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are linear projections of the input tokens, and d_k is the dimension of the key vectors. In the MHSA mechanism, this is performed H times in parallel, resulting in a full attention tensor $\mathbf{A}$.

Visual Token Pruning Problem. The large number of visual tokens, N , from the vision encoder presents a primary bottleneck for VLM inference. The objective of visual token pruning is therefore to select a much smaller, yet representative, subset of tokens. Formally, we aim to learn a selection function, S , that generates a set of token indices $\mathcal{J} \subset \{1, \dots, N\}$. The pruned token set is then denoted as $\mathcal{X}' = \{x_i \mid i \in \mathcal{J}\}$. Our goal is to find an optimal index set $\mathcal{J}$ that minimizes the performance degradation caused by pruning, subject to a budget constraint $|\mathcal{J}| = K$, where $K \ll N$. This optimization problem is formulated as:

$$\min_{\mathcal{J}} \mathcal{L}(f(\mathcal{X}), f(\mathcal{X}')) \quad \text{subject to} \quad |\mathcal{J}| = K. \quad (2)$$

Here, f denotes the end-to-end VLM mapping function, and $\mathcal{L}$ is the task-specific loss function. Our primary con-

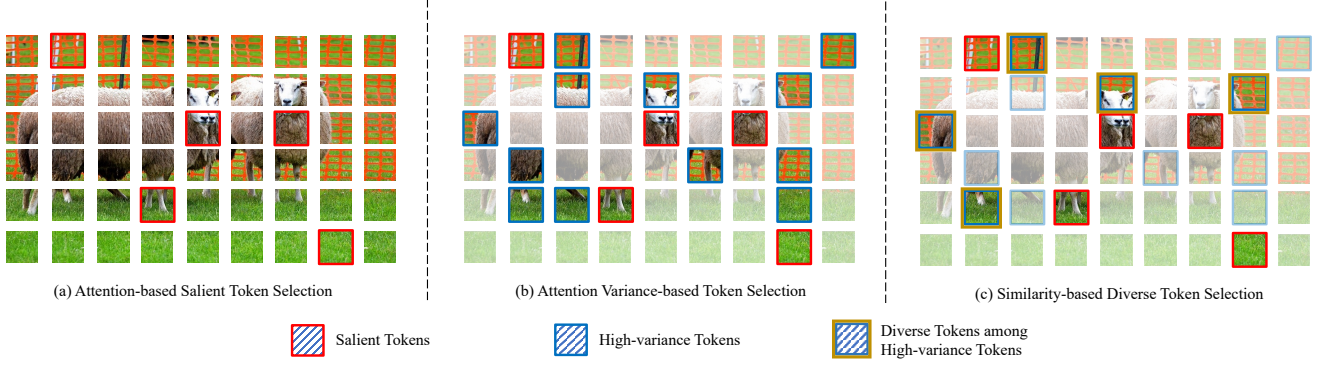


Figure 3. **The overview of PrismPrune.** The process includes (a) attention-based salient token selection, (b) attention variance-based token selection, and (c) similarity-based diverse token selection.

tribution is the design of a novel selection function S that efficiently generates a high-quality index set $\mathcal{J}$.

3.2. PrismPrune

As shown in Figure 3, rather than relying on a singular statistic, our token pruning methodology decomposes the attention signal to isolate two orthogonal properties. This allows us to propose a two-stage pruning framework: first, preserving high-saliency tokens based on mean attention, and second, supplementing with high-diversity tokens identified through attention variance. The overall process is outlined in Algorithm 1.

Saliency-based Selection via Mean Attention. The first stage aims to ensure that visual tokens constituting the core content are preferentially retained. We leverage attention scores from the vision encoder as a proxy for token saliency. Specifically, for each attention head $h \in \{1, \dots, H\}$, we extract the row vector from the attention map $\mathbf{A}_h$ corresponding to the class token (CLS), denoted as $\mathbf{a}_h^{\text{cls}}$. This vector represents the attention from the CLS token to all other visual tokens within that head. We then compute the mean of these vectors across all heads to obtain a single saliency score vector s_{sal} :

$$s_{\text{sal}} = \frac{1}{H} \sum_{h=1}^H \mathbf{a}_h^{\text{cls}}. \quad (3)$$

Based on these scores, we select the indices of the top- K_{sal} tokens to form the saliency index set $\mathcal{J}_{\text{sal}}$:

$$\mathcal{J}_{\text{sal}} = \text{top-k}(s_{\text{sal}}, K_{\text{sal}}), \quad (4)$$

where the $\text{top-k}(\cdot)$ function returns the indices of the K_{sal} highest-scoring elements.

Diversity-aware Supplementation via Attention Variance. Relying solely on high-saliency tokens often leads to the loss of contextual information, which is crucial for

Algorithm 2 Farthest Point Sampling

```
# Function FarthestPointSampling(P, D)
# P: Candidate points, shape (M, C)
# D: Number of points to select

# Compute point distances
dist_matrix = Distance(P) # Shape (M, M)

# Initialize with the most isolated point
first_idx = dist_matrix.mean(dim=-1).argmax()
s_idx = [first_idx]
min_dists = dist_matrix[first_idx]

# Iteratively add the farthest point to the
# selected set
for _ in range(1, D):
    next_idx = min_dists.argmax()
    s_idx.append(next_idx)

# Update min distances to the expanding
# selected set
new_dists = dist_matrix[next_idx]
min_dists = minimum(min_dists, new_dists)
```

complex visual reasoning. To mitigate this issue, our second stage selects an information-rich supplementary set of indices, $\mathcal{J}_{\text{div}}$, from the remaining token pool. This is accomplished using our efficient “filter-then-sample” strategy.

Our approach finds its theoretical grounding in the unified theory of ensemble learning [46], which establishes a formal mathematical link between a system’s diversity and the variance among its members’ predictions. When conceptualizing multi-head attention as an intrinsic ensemble where each head is a member, this theory justifies using variance as a direct proxy for diversity. Consequently, a token exhibiting high variance across different attention heads is a strong indicator of high diversity. Such a token is likely to be interpreted from multiple, diverse semantic perspectives, making it an information-dense and robust feature. An extended analysis is in the supplementary material.

Method	MME	TextVQA	SQA	MM-Vet	MMB-CN	GQA	VQA ^{v2}	MMBench	POPE	VizWiz	Avg. Rel (%)
<i>Vanilla, 576 Tokens (100%)</i>											
LLaVA-1.5-7B	1510.7	58.2	66.8	31.1	58.3	62.0	78.5	64.3	85.9	50.0	100.0%
<i>Retain 128 Tokens (↓ 77.8%)</i>											
FastV[8]	1208.9	50.6	60.2	28.1	51.4	49.6	61.8	56.1	59.6	51.3	85.4%
SparseVLM[58]	1376.2	54.9	67.1	30.0	51.1	56.0	73.8	60.0	80.5	51.4	94.4%
DART[45]	-	56.4	69.1	-	57.5	58.7	75.9	63.2	80.1	51.7	98.2%
VisionZip[50]	1432.4	56.8	68.9	32.6	56.7	57.6	75.6	62.0	83.2	52.0	98.4%
VisPruner[56]	1461.4	57.0	69.1	33.7	57.3	58.2	75.8	62.7	84.6	52.7	99.6%
PrismPrune	1456.2	57.3	69.3	33.8	57.6	58.8	76.2	62.7	85.4	52.9	100.1%
PrismPrune [†]	1481.6	57.4	68.0	34.7	57.7	60.9	76.4	63.8	84.2	53.0	100.8%
<i>Retain 64 Tokens (↓ 88.9%)</i>											
FastV[8]	1019.6	47.8	51.1	25.8	42.7	46.1	55.0	48.0	48.0	50.8	75.9%
PruMerge+[37]	1198.2	53.0	68.6	25.9	51.0	54.9	67.4	59.3	77.4	52.9	90.6%
DART[45]	-	54.4	69.8	-	53.2	55.9	72.4	60.6	73.9	51.6	94.4%
VisionZip[50]	1365.6	55.5	69.0	31.7	55.4	55.1	72.4	60.1	77.0	52.9	95.6%
VisPruner[56]	1369.9	55.8	69.1	32.3	55.1	55.4	72.7	61.3	80.4	53.3	96.6%
PrismPrune	1387.5	55.9	69.2	32.3	55.3	56.5	74.2	61.4	82.1	53.7	97.5%
PrismPrune [†]	1424.1	56.0	67.7	32.7	56.5	59.0	74.7	62.5	83.4	53.0	98.4%
<i>Retain 32 Tokens (↓ 94.4%)</i>											
FastV[8]	884.6	42.5	42.6	20.7	33.2	41.5	43.4	37.8	32.5	51.7	64.1%
SparseVLM[58]	1046.7	46.1	57.3	18.6	40.6	48.3	58.6	51.4	67.9	51.9	77.9%
PruMerge+[37]	940.8	50.6	68.5	21.4	47.0	51.1	54.9	56.8	70.9	52.8	83.0%
VisionZip[50]	1247.4	53.1	68.8	25.5	50.3	51.8	67.1	57.7	68.7	52.9	89.0%
VisPruner[56]	1271.0	53.9	69.2	28.8	52.7	52.2	67.7	58.4	72.7	53.0	91.9%
PrismPrune	1331.3	54.1	69.2	28.8	52.8	53.9	70.3	59.1	76.1	53.3	93.1%
PrismPrune [†]	1386.6	55.0	68.0	29.1	52.3	56.0	70.3	60.7	80.3	53.2	94.5%

Table 1. **Performance comparison of PrismPrune on LLaVA-1.5-7B against baselines.** The vanilla number of visual tokens is 576. PrismPrune[†] indicates our model after a lightweight fine-tuning step. The final column shows the average performance relative to the original model, with the best result in each section highlighted in **red**.

Therefore, we introduce attention variance to guide the diversity-aware selection. For each visual token i , we compute the variance of its attention scores from the CLS token across the set of head-specific vectors $\{\mathbf{a}_1^{\text{cls}}, \dots, \mathbf{a}_H^{\text{cls}}\}$. This yields a variance score vector s_{var} :

$$s_{\text{var},i} = \text{Var}(\{\mathbf{a}_{h,i}^{\text{cls}} \mid h = 1, \dots, H\}). \quad (5)$$

The selection then proceeds in two steps. First, we filter for a high-quality candidate pool, $\mathcal{X}_{\text{pool}}$, composed of the M tokens with the highest variance scores from the unelected set, where $M \geq K_{\text{div}}$. Subsequently, inspired by DivPrune[2], we execute a greedy iterative selection algorithm on $\mathcal{X}_{\text{pool}}$, which is analogous to Farthest Point Sampling (FPS)[13]. This process initializes the selection set and, at each iteration, adds the token from the pool that is farthest in feature space from the currently selected set, until K_{div} tokens are chosen, as detailed in Algorithm 2. This yields the diversity index set $\mathcal{J}_{\text{div}}$.

Finally, the complete set of selected indices is the union of the outputs from both stages, $\mathcal{J} = \mathcal{J}_{\text{sal}} \cup \mathcal{J}_{\text{div}}$, with a total size of $K = K_{\text{sal}} + K_{\text{div}}$. The pruned set of tokens is

then constructed as $\mathcal{X}' = \{x_i \mid i \in \mathcal{J}\}$. This hybrid strategy, combining mean and variance, ensures that the pruned token set preserves the fidelity of the core content and maintains the representational breadth of the global context.

3.3. Parameter-efficient Fine-tuning

While our pruning methodology is inherently training-free, it introduces a structural modification at the interface between the vision encoder and the LLM: the sequence of visual tokens fed to the language model is significantly compressed[50]. This alteration can induce a subtle distributional shift. This is because the VLM was originally pre-trained and instruction-tuned on the complete set of visual tokens. To counteract this shift and recalibrate the modality interface, we incorporate a parameter-efficient fine-tuning (PEFT) stage. We only finetune the multimodal projection layer responsible for this transformation. During this phase, the parameters of the computationally intensive vision encoder and the LLM remain completely frozen, rendering the entire fine-tuning process exceptionally lightweight.

Specifically, we leverage a fraction of the original

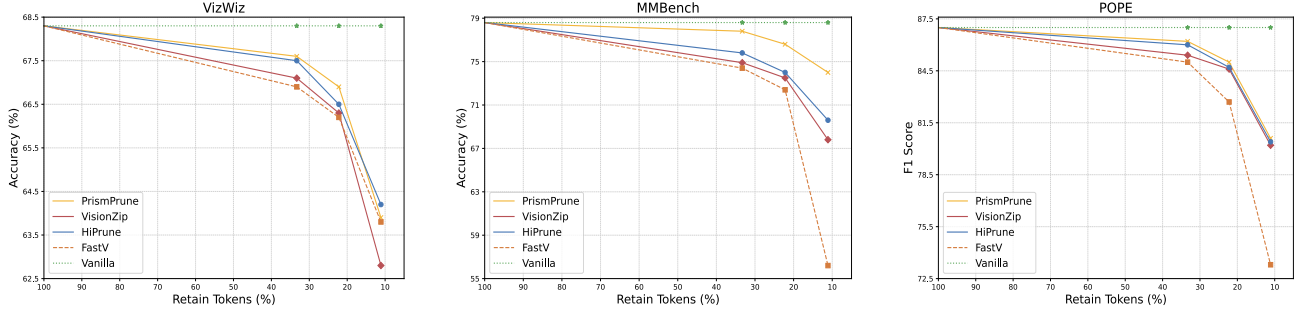


Figure 4. Performance of PrismPrune on the Qwen2.5-VL-3B-Instruct.

instruction-tuning data to update the projector’s parameters. The efficiency of this stage is empirically validated in our experiments: for the LLaVA-1.5-7B model, the training takes around 2 hours on 8 NVIDIA A100 GPUs, requiring only one-fifth of the original fine-tuning time.

4. Experiment

In this section, we conduct extensive experiments to evaluate our proposed method by applying it to a variety of VLMs across diverse tasks, including both image and video understanding. We compare our approach against several existing state-of-the-art methods, perform meticulous ablation studies to validate our design choices, and provide a thorough analysis of computational efficiency.

4.1. Results on Image Understanding

Datasets and model architectures. We conduct our evaluation on 10 widely-used benchmarks, encompassing conventional Visual Question Answering (VQA) datasets such as VQAv2[14], GQA[19], VizWiz[15], ScienceQA[33], and TextVQA[38], as well as other prominent multimodal benchmarks including POPE[24], MME[7], MMBench[32], MMBench-CN[32], and MMVet[53]. We apply our method to a variety of VLM architectures, including LLaVA-1.5[29] for standard-resolution inputs, LLaVA-NeXT[30] for high-resolution imagery, and other prevalent models such as Qwen2.5-VL[5].

Results on LLaVA-1.5. As presented in Table 1, we apply our proposed method to LLaVA-1.5 to demonstrate its performance on image understanding tasks. PrismPrune denotes the direct application of our method at inference time, while PrismPrune[†] represents the results after our lightweight fine-tuning. For a comprehensive comparison, we report performance as a relative percentage, where the accuracy of the original, unpruned model serves as the 100% upper bound. Following the prior work[56], we evaluate the efficacy of PrismPrune under three visual token budgets (128, 64, and 32).

When the number of visual tokens is reduced from 576

Method	MME	TextVQA	POPE	GQA	VQA ^{v2}	Rel
<i>Vanilla, 2880 Tokens (100%)</i>						
LLaVA-1.6-7B	1513.8	59.6	86.3	62.9	81.2	100%
<i>Retain 640 Tokens(↓ 77.8%)</i>						
FastV[8]	1477.3	58.4	83.1	60.4	78.9	97.0%
SparseVLM[58]	1456.3	56.2	80.9	59.1	78.2	94.9%
VisionZip[50]	1468.4	58.5	82.2	60.1	79.2	96.7%
VisPruner[56]	1480.7	59.3	85.9	61.6	79.8	98.6%
PrismPrune	1492.1	60.4	86.4	61.7	79.2	99.2%
<i>Retain 320 Tokens(↓ 88.9%)</i>						
FastV[8]	1282.9	55.7	71.7	55.9	71.9	87.7%
SparseVLM[58]	1342.7	52.4	73.5	56.5	71.4	87.9%
VisionZip[50]	1348.8	55.3	75.0	58.1	74.2	90.5%
VisPruner[56]	1370.1	57.6	80.4	58.4	75.7	93.3%
PrismPrune	1432.6	58.9	83.3	59.9	77.0	96.0%
<i>Retain 160 Tokens(↓ 94.4%)</i>						
FastV[8]	1079.5	51.9	51.7	49.8	61.8	74.7%
SparseVLM[58]	1167.1	45.1	54.6	50.2	62.2	74.5%
VisionZip[50]	1239.7	54.7	59.4	54.3	67.3	82.3%
VisPruner[56]	1226.0	56.0	72.9	54.7	70.6	86.7%
PrismPrune	1315.8	57.2	78.2	57.5	74.1	91.2%

Table 2. Performance comparison of different pruning methods on LLaVA-1.6-7B. The vanilla number of visual tokens is 2880. Rel represents the average percentage of performance maintained at the corresponding token retention level.

to 128, PrismPrune maintains 100% of the original average accuracy without requiring any additional training, outperforming FastV and VisionZip by 14.7% and 1.7%, respectively. Moreover, when pruned aggressively to only 32 tokens, our method surpasses FastV and VisionZip by even more significant margins of 29.0% and 4.1% in average accuracy. Additionally, our PrismPrune[†] delivers further performance improvements. By retaining only 5% of the original visual tokens, this efficient finetuning approach elevates the average accuracy to 94.5%, incurring only a minor 5.5% drop compared to the original model, which utilizes 20 times more visual tokens.

Results on LLaVA-NeXT. Leveraging higher-resolution inputs effectively enhances VLM performance on various

Method	TGIF-QA		MSVD-QA		MSRVTT-QA		Average	
	Acc.	Score	Acc.	Score	Acc.	Score	Acc.	Score
<i>Vanilla, 2048 Tokens (100%)</i>								
Video-LLaVA	18.9	2.53	71.7	3.95	57.5	3.50	49.4	3.33
<i>Retain 455 Tokens(↓ 77.8%)</i>								
FastV[8]	19.2	2.50	69.1	3.91	54.4	3.42	47.6	3.28
VisPruner[56]	18.4	2.49	70.2	3.95	56.7	3.50	48.4	3.31
PrismPrune	18.0	2.49	71.4	3.96	56.9	3.47	48.8	3.31
<i>Retain 227 Tokens(↓ 88.9%)</i>								
FastV[8]	14.3	2.42	68.9	3.90	53.0	3.40	45.4	3.24
VisPruner[56]	15.9	2.41	69.3	3.92	55.6	3.45	46.9	3.26
PrismPrune	15.8	2.44	69.4	3.93	55.7	3.44	47.0	3.27
<i>Retain 114 Tokens(↓ 94.4%)</i>								
FastV[8]	10.6	2.29	64.1	3.78	52.4	3.39	42.4	3.15
VisPruner[56]	14.1	2.35	65.4	3.79	54.1	3.41	44.5	3.18
PrismPrune	14.5	2.37	66.8	3.86	55.1	3.43	45.5	3.22

Table 3. **Performance comparison of different pruning methods on Video-LLaVA-7B.** Performance was assessed using the initial 1,000 samples from each benchmark, with gpt-3.5-turbo employed for automated scoring.

tasks. However, higher resolution leads to a substantial increase in visual tokens, which escalates the computational overhead of VLMs. To further validate the efficacy of PrismPrune, we apply it to a VLM adapted for high-resolution inputs, LLaVA-NeXT, which can process up to 2,880 visual tokens. We establish three token budget settings (640, 320, and 160) to assess our method’s advantages. As shown in Table 2, PrismPrune achieves superior performance across all three settings. Notably, while retaining 320 tokens, our method reaches 96.0% of the original average accuracy, marking an improvement of 5.5% and 2.7% over VisionZip and VisPruner, respectively. Even when pruned to 160 tokens, our method sustains a strong 91.2% average accuracy, outperforming VisionZip and VisPruner by substantial margins of 8.9% and 4.5%, respectively.

Results on Qwen2.5-VL. To validate the versatility of PrismPrune, we integrated it into the Qwen2.5-VL architecture. This is a crucial test, as many competing approaches that rely on specific components like a CLIP[36] text encoder or special tokens are often incompatible with such models. As illustrated in Figure 4, PrismPrune’s superiority becomes increasingly evident as the number of visual tokens is reduced across the POPE, MMBench, and VizWiz benchmarks. Notably, under aggressive compression, the performance gap between PrismPrune and the previous methods widens dramatically on the MMBench dataset. These results confirm the effectiveness and adaptability of our method across diverse model architectures.

4.2. Results on Video Understanding

Datasets and Model architectures. We evaluate our proposed method on three video question answering bench-

Method	Token	Storage (MB)	GPU Memory (GB)	CUDA Time (ms)
LLaVA-NeXT-7B	2880	1440	17.0	313
FastV[8]	640	380	16.9	148
PrismPrune	640	351	14.9	115

Table 4. **Resource efficiency analysis with LLaVA-NeXT-7B on POPE.** We report the number of visual tokens, storage size, peak GPU memory usage, and single-pass CUDA time.

Method	Token	FLOPS ↓	Total Time ↓	Δ	Prefilling Time ↓	Δ
LLaVA-NeXT-7B	2880	43.6T	2293s	—	218ms	—
FastV[8]	160	6.3T	1792s	1.3×	119ms	1.8×
PrismPrune	160	3.4T	791s	2.9×	21.4ms	10.2×

Table 5. **Comparison of inference speed and computational cost on LLaVA-NeXT-7B.** We report total FLOPS, total generation time, and prefilling time. The speed-up ratio (Δ) is calculated against the unpruned baseline.

marks: TGIF-QA[20], MSVD-QA[48], and MSRVTT-QA[48]. Our approach is applied to the Video-LLaVA[26] model. In line with prior works[26, 34], we employ a GPT-assisted evaluation protocol where scores assigned by ChatGPT serve as the primary performance metric.

Results on Video-LLaVA. Video understanding is another scenario with high visual redundancy. Due to the commercial API usage limits, we follow[8] to use the first 1K samples from each benchmark in our experiments. The evaluation results are presented in Table 3. As a baseline, the Video-LLaVA model processes 8 video frames at a 224x224 resolution, corresponding to 2048 visual tokens. At a high compression rate (retaining just 455 tokens, or 22.2%), PrismPrune maintains 98.7% of the original performance. Even with only 114 tokens retained (a 94.4% reduction), it still achieves 92.1% of the original performance. This significantly surpasses competing methods like FastV (86%) and VisPruner (90.1%), highlighting PrismPrune’s efficacy for continuous video inputs.

4.3. Efficiency Analysis

To assess the computational efficiency of PrismPrune, we follow established practices and conduct a detailed analysis using a suite of metrics, including total inference time, pre-filling time, FLOPS, cache storage, GPU memory, and CUDA latency. Our experiments were conducted on the LLaVA-NeXT-7B model, benchmarking against FastV[8] as a representative baseline. All experiments were performed on the POPE using a single NVIDIA A100 GPU.

As detailed in Table 4, when retaining 640 tokens, PrismPrune achieves a 2.7-fold reduction in CUDA latency. At a comparable token reduction ratio, PrismPrune demonstrates superior resource utilization, requiring less cache storage

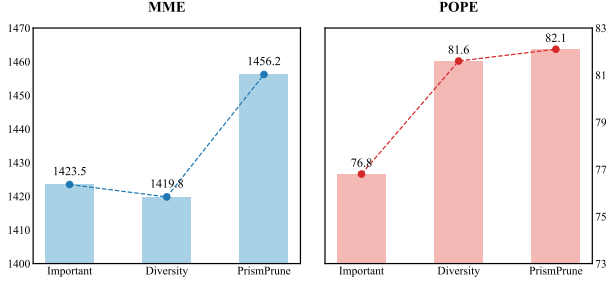


Figure 5. **Ablation study of the core components in our method.** Important denotes selecting tokens based solely on high attention scores. Diversity uses farthest point sampling to select visual tokens. PrismPrune is our full method, which incorporates an additional variance-filtering step for token selection.

and GPU memory than FastV. Furthermore, in terms of inference and pre-filling latency, PrismPrune not only delivers state-of-the-art performance but also achieves substantial efficiency gains. Specifically, as shown in Table 5, our method reduces FLOPS by a factor of 12.8, lowers the total inference time by 2.9 $\times$, and slashes the pre-filling time by an impressive 10.2 $\times$ on the POPE. Crucially, PrismPrune completes the pruning process entirely within the vision encoder. This design choice enables full compatibility with modern, optimized attention mechanisms such as FlashAttention. This is a distinct advantage not afforded to methods that depend on text-vision attention scores, which often create bottlenecks and are incompatible with such frameworks.

4.4. Ablation Study

To validate the contribution of each component within PrismPrune, we conducted a series of ablation studies on the LLaVA-1.5-7B model, with results benchmarked on the POPE and MME datasets as illustrated in Figure 5. We found that relying solely on attention scores for token retention leads to a significant loss of information from the background. Conversely, applying a global diversity algorithm directly overlooks the critical features captured by high-variance tokens. These findings substantiate the design of PrismPrune, demonstrating that our novel variance-filtering step is essential for efficiently curating a high-potential candidate pool upon which diversity sampling can be effectively performed.

To further validate the efficiency and effectiveness of our “filter-then-sample” strategy, we conduct an ablation study on LLaVA-NeXT-7B, with results shown in Table 6. We compare the full PrismPrune against two variants. The first, Global FPS, ablates our variance-filtering step by applying Farthest Point Sampling directly to all remaining tokens after the initial saliency selection, a strategy similar to that used in DivPrune [2]. The second variant, Bipartite Pruning, replaces FPS with a fast pairing-based heuris-

Method	MME			POPE		
	Total Time	Δ	score	Total Time	Δ	score
<i>Vanilla, 2880 Tokens (100%)</i>						
LLaVA-NeXT-7B	820s	—	1513.8	2293s	—	86.3
<i>Retain 320 Tokens ($\downarrow$ 88.9%)</i>						
Bipartite Pruning	287s	2.9 $\times$	1370.1	923s	2.5 $\times$	80.4
Global FPS	257s	3.2 $\times$	1403.0	908s	2.5 $\times$	82.5
PrismPrune	233s	3.5$\times$	1432.6	853s	2.7$\times$	83.3
<i>Retain 160 Tokens ($\downarrow$ 94.4%)</i>						
Bipartite Pruning	258s	3.2 $\times$	1226.0	865s	2.7 $\times$	72.9
Global FPS	217s	3.8 $\times$	1297.7	819s	2.8 $\times$	77.0
PrismPrune	209s	3.9$\times$	1315.8	791s	2.9$\times$	78.2

Table 6. **Ablation study on diversity selection strategies.** PrismPrune, with its variance-based filtering, is both the fastest and highest-performing method, validating its superior efficiency-performance trade-off.

tic for diversity, conceptually similar to the method in VisPruner [56]. The results are conclusive: Global FPS is significantly slower than our method while achieving lower performance. Conversely, Bipartite Pruning suffers a substantial performance drop, proving that a principled diversity algorithm is necessary. These findings confirm that PrismPrune’s design strikes an optimal trade-off.

5. Conclusion

In this paper, we addressed the critical challenge of efficient visual token pruning in VLMs. We identified that the central challenge lies in efficiently and accurately defining and identifying the set of important tokens, a task where previous approaches have struggled. Our work introduces PrismPrune, a novel framework founded on the key insight that the signals for saliency and diversity are already intrinsically decoupled within the multi-head attention mechanism. Instead of attempting to balance these competing objectives, PrismPrune leverages the distinct statistical properties of attention in a principled, two-stage process. First, it preserves a core set of salient tokens by selecting those with the highest mean attention scores. Subsequently, it addresses diversity via an efficient “filter-then-sample” strategy guided by our key contribution: attention variance. Grounded in the unified theory of ensemble learning, this theoretically-justified measure is used to filter the remaining tokens into a candidate pool. We then perform diversity-aware sampling exclusively on this curated subset, drastically reducing computational overhead while ensuring a representative selection. Furthermore, we demonstrate that a lightweight fine-tuning step can mitigate the subtle modality misalignment introduced by pruning, yielding superior results. Extensive experiments validate our approach, showing that PrismPrune significantly outperforms existing methods in both performance and efficiency.

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